

Requirement Analysis

Technology Stack (Architecture & Stack)

Date	21 June 2026
Team ID	
Project Name	Optimizing User, Group, and Role Management with Access Control and Workflows
Maximum Marks	4 Marks

Technical Architecture:

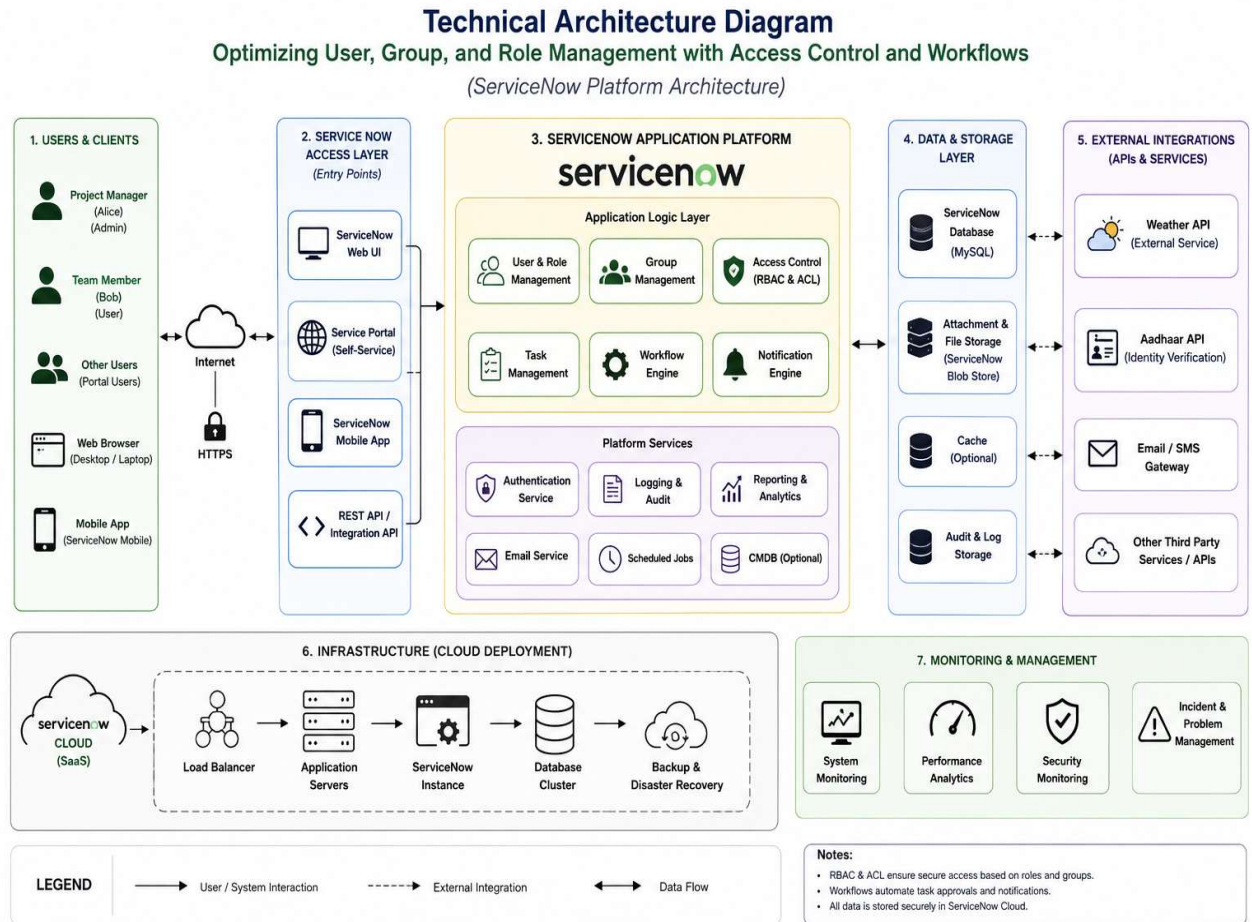


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web interface used by administrators, Project Manager, and Team Member to access the application	ServiceNow Web UI
2	Application Logic – User & Role Management	Handles user creation, group management, and role assignment	ServiceNow User Administration
3	Application Logic – Access Control	Implements Role-Based Access Control (RBAC) and Access Control Lists (ACLs)	ServiceNow ACL, RBAC
4	Application Logic – Workflow Automation	Automates task assignment, status updates, and approval processes	ServiceNow Flow Designer
5	Database	Stores users, groups, roles, projects, tasks, and workflow records	ServiceNow Database (MySQL-based platform database)
6	Cloud Database	Cloud-hosted managed database provided by the ServiceNow platform	ServiceNow Cloud Platform
7	File Storage	Stores attachments, documents, and task-related files	ServiceNow Attachment Storage
8	External API – Notifications	Sends workflow notifications and approval alerts	ServiceNow Notification Service (Email)
9	External API – Authentication	User authentication and identity verification	ServiceNow Authentication / Identity Management
10	Security Model	Controls user permissions, record-level access, and approval workflows	RBAC, ACL, User Criteria
11	Infrastructure (Server / Cloud)	Cloud-hosted enterprise ServiceNow instance	ServiceNow SaaS Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Platform-based development using low-code/no-code components	ServiceNow App Engine & Flow Designer
2	Security Implementations	Role-Based Access Control, Access Control Lists, authentication, authorization, audit logging	RBAC, ACL, ServiceNow Security Framework
3	Scalable Architecture	Cloud-native, multi-tier architecture supporting multiple users and applications	ServiceNow Cloud Architecture
4	Availability	High availability with enterprise cloud infrastructure and automatic backup	ServiceNow SaaS Platform
5	Performance	Optimized workflows, indexed tables, automated processing, and cloud-managed infrastructure	Flow Designer, ServiceNow Platform Optimization